MVP Scope Document — Smart Irrigation Alert Dashboard

Project Title

Jenkins-Based Smart Irrigation Alert Dashboard

Student Details

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- **Academic Year:** 2026-27 ODD Semester

Scope Summary

This MVP focuses on delivering a functional web-based dashboard that simulates IoT sensor data for irrigation monitoring and generates alerts when moisture levels are critically low. The entire project is built and deployed using an end-to-end DevOps pipeline.

In-Scope Features (MVP)

#	Feature	Description
1	Data/Event Entry	Form/API to add new sensor zone data (moisture %, temperature, zone name)
2	Searchable Dashboard	Table/grid view of all sensors with search and filter capabilities
3	Summary Indicators	Cards showing average moisture, total zones, active alerts count
4	Status Drill-Down	Click on any zone to view detailed sensor readings and history
5	Alert/Exception View	Zones below 30% moisture are highlighted with alert badges

Out-of-Scope (Future Enhancements)

- Real IoT hardware integration (physical sensors)
- Mobile application (Android/iOS)
- SMS/Email notification system
- Machine learning-based irrigation prediction
- Multi-tenant support for multiple farms
- Weather API integration

DevOps Pipeline Scope

Stage	**Tool**
Version Control	Git + GitHub
CI/CD	Jenkins (Pipeline as Code)
Build	Maven
Automated Testing	Selenium WebDriver
Containerization	Docker
Config Management	Ansible
Deployment	Docker Container (via Jenkins)

Approval

This MVP scope has been reviewed and frozen for the DevOps project assignment.